GATE ME 90 Days Challenge

Day 54/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	Week 9 Revision	49 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed revision of the week: IC engine cycles, combustion and performance, refrigeration and air-conditioning, complex integration and numerical methods, steam and hydraulic turbines, and the iron–carbon diagram and heat treatment of steels. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Detonation (knock) in a spark-ignition engine is caused by

(A) auto-ignition of the end gas ahead of the advancing flame front

(B) a long ignition delay of the fuel injected

(C) pre-ignition of the charge by the spark plug before the spark occurs

(D) incomplete combustion due to a lean mixture

Q2 NAT

In a trial on a diesel engine, fuel (calorific value 42 000 kJ/kg) is burnt at 8 kg/h, and cooling water flows at 0.2 kg/s with a temperature rise of 15 °C (c = 4.18 kJ/kg·K). The percentage of the fuel energy carried away by the cooling water (round off to two decimal places) is _____

Q3 NAT

An air-standard Diesel cycle (γ = 1.4) has a compression ratio of 16, and the fuel is cut off at 6 % of the stroke. Its thermal efficiency (in %, round off to one decimal place) is _____

Q4 MCQ

In a lithium bromide–water vapour-absorption refrigeration system,

(A) lithium bromide is the refrigerant and water the absorbent

(B) water is the refrigerant, so the system cannot produce temperatures below about 0 °C

(C) ammonia is added as the refrigerant

(D) a mechanical compressor raises the pressure of the refrigerant vapour

Q5 NAT

A 5 TR ammonia refrigerator (1 TR = 3.517 kW) has a refrigerant enthalpy of 100 kJ/kg at evaporator inlet and 245 kJ/kg at evaporator exit. The specific volume of the vapour at compressor suction is 0.5 m^3/kg and the volumetric efficiency of the compressor is 80 %. The swept-volume rate needed (in m^3/s, round off to four decimal places) is _____

Q6 MCQ

A room has a sensible heat load of 25 kW and a latent heat load of 5 kW. Its room sensible heat factor (RSHF), and the effect of a rise in occupancy that adds mainly latent load, are

(A) 0.83; the RSHF rises and the supply air may be more humid

(B) 0.83; the RSHF falls and the supply air must be drier

(C) 0.20; the RSHF falls

(D) 5.0; the RSHF is unaffected

Q7 MCQ

Let C be the circle $|z| = 1$ traversed anticlockwise. The value of $\oint_C dz/(z^2 + 4)$ is

(A) $\pi i/2$

(B) $\pi/2$

(C) $2\pi i$

(D) 0

Q8 NAT

The root of $x = \cos x$ (x in radians) is sought by the Newton–Raphson method starting from $x_0 = 0.5$. The value of x_1 (round off to four decimal places) is _____

Q9 NAT

The integral of x^4 from $x = 0$ to $x = 3$ is evaluated by Simpson's 3/8 rule with three equal strips ($h = 1$). The value obtained (round off to one decimal place) is _____

Q10 MCQ

The hydraulic turbine whose runner blades can be rotated about their own axes while the turbine is running, giving a high efficiency at part load, is the

(A) Pelton wheel

(B) Francis turbine

(C) Kaplan turbine

(D) propeller turbine with fixed blades

Q11 NAT

The isentropic enthalpy drop in the nozzle of an impulse steam turbine is 180 kJ/kg and the nozzle efficiency is 92 %. Neglecting the velocity of approach, the steam velocity at nozzle exit (in m/s, round off to one decimal place) is _____

Q12 MCQ

For a given head and power, choosing a turbine of higher specific speed means that

(A) the turbine runs faster, so the turbine and generator are smaller and cheaper

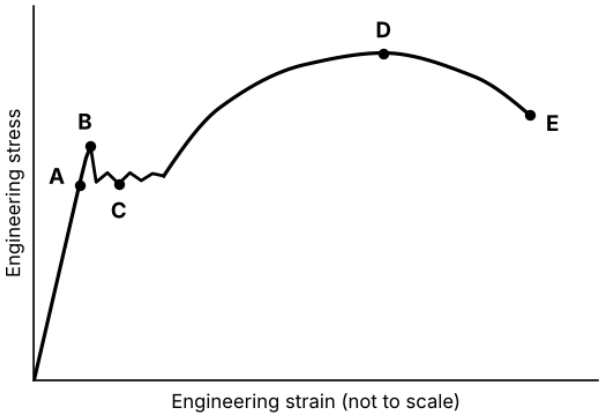
(B) the turbine runs slower and needs a larger generator

(C) the turbine must be of the Pelton type

(D) the risk of cavitation is reduced

Q13 MCQ

The engineering stress–strain curve of mild steel is shown with points A to E. The lower yield point is



(A) B
(B) D
(C) E
(D) C

Q14 NAT

A plain carbon steel with 1.0 % C is slowly cooled to room temperature. With the eutectoid at 0.76 % C and cementite at 6.67 % C, the mass percentage of pearlite in its microstructure (round off to two decimal places) is _____

Q15 MCQ

The amount of retained austenite in a hardened high-carbon steel can be reduced by

(A) tempering at 150 °C only
(B) quenching from a higher austenitising temperature
(C) sub-zero (cryogenic) treatment immediately after quenching
(D) adding more nickel or manganese

Answer Key

Q	Type	Answer
Q1	MCQ	(A) auto-ignition of the end gas ahead of the advancing flame front
Q2	NAT	13.44 (accept 13.31 to 13.57)
Q3	NAT	61.9 (accept 61.5 to 62.3)
Q4	MCQ	(B) water is the refrigerant, so the system cannot produce temperatures below about 0 °C
Q5	NAT	0.0758 (accept 0.0754 to 0.0762)
Q6	MCQ	(B) 0.83; the RSHF falls and the supply air must be drier
Q7	MCQ	(D) 0
Q8	NAT	0.7552 (accept 0.7550 to 0.7554)
Q9	NAT	49.5 (accept 49.4 to 49.6)
Q10	MCQ	(C) Kaplan turbine
Q11	NAT	575.5 (accept 572.6 to 578.4)
Q12	MCQ	(A) the turbine runs faster, so the turbine and generator are smaller and cheaper
Q13	MCQ	(D) C
Q14	NAT	95.94 (accept 95.45 to 96.43)
Q15	MCQ	(C) sub-zero (cryogenic) treatment immediately after quenching

Detailed Solutions

Q1 **Answer: (A) auto-ignition of the end gas ahead of the advancing flame front**

Concept: In an SI engine the flame travels from the spark plug across the chamber, compressing the unburnt end gas ahead of it.

If the end gas reaches its self-ignition temperature before the flame arrives, it auto-ignites almost instantaneously, producing pressure waves and the knocking sound: **(A)**.

A long ignition delay is the cause of knock in CI (diesel) engines; pre-ignition by hot spots is a different phenomenon.

Exam tip: SI knock is reduced by high-octane fuel, retarded spark, lower compression ratio and cooler charge — the opposite of what reduces CI knock.

Q2 **Answer: 13.44 (accept 13.31 to 13.57)**

Concept: Heat balance sheet: each item is expressed as a percentage of the fuel energy supplied.

Fuel energy = 8 × 42 000/3600 = 93.33 kW

Cooling water = 0.2 × 4.18 × 15 = 12.54 kW, so the share = 12.54/93.33 = **13.44 %**

Exam tip: A typical diesel engine heat balance: brake power 30–35 %, cooling water 25–30 %, exhaust 25–35 %, unaccounted the rest.

Q3 Answer: 61.9 (accept 61.5 to 62.3)

Concept: Cut-off at k % of the stroke: $V_3 = V_c + k\,V_s$, so $\rho = 1 + k(r - 1)$.

$\rho = 1 + 0.06 \times 15 = 1.9$

$\eta = 1 - (1/r^{\gamma-1})(\rho^{\gamma} - 1)/[\gamma(\rho - 1)] = 1 - (1/3.031)(2.456 - 1)/(1.4 \times 0.9) = 1 - 0.3299 \times 1.1557$ = **61.9 %**

Exam tip: Do not take $\rho = 0.06$ or 1.06; the cut-off percentage refers to the swept volume, not the clearance volume.

Q4 Answer: (B) water is the refrigerant, so the system cannot produce temperatures below about 0 °C

Concept: In absorption systems the refrigerant is the volatile component and the absorbent takes it up in the absorber.

In LiBr–water units, water evaporates at very low pressure in the evaporator (refrigerant) and LiBr solution absorbs it; water freezes at 0 °C, so these units are used for air-conditioning (chilled water at 5–10 °C): **(B)**.

In the NH_3–water system ammonia is the refrigerant and water the absorbent, so sub-zero temperatures are possible.

Exam tip: The generator, absorber and pump replace the compressor; the input is mainly heat, not work.

Q5 Answer: 0.0758 (accept 0.0754 to 0.0762)

Concept: Mass flow = capacity/refrigerating effect; actual suction volume = $\dot{m}v$; swept volume = actual volume/η_v.

$\dot{m} = 5 \times 3.517/(245 - 100) = 17.585/145 = 0.1213$ kg/s

Swept volume rate = $0.1213 \times 0.5/0.8$ = **0.0758 m³/s**

Exam tip: Lower evaporator pressure → larger suction specific volume → a bigger compressor for the same capacity.

Q6 Answer: (B) 0.83; the RSHF falls and the supply air must be drier

Concept: RSHF = room sensible heat/(sensible + latent heat).

RSHF = 25/(25 + 5) = 0.83.

More latent (moisture) load makes the RSHF smaller; to absorb the extra moisture the supply air must have a lower specific humidity, and the RSHF line on the psychrometric chart becomes steeper: **(B)**.

Exam tip: Auditoriums and kitchens (many people, cooking) have a low RSHF; computer rooms have an RSHF close to 1.

Q7 Answer: (D) 0

Concept: Cauchy's theorem: the integral of a function analytic on and inside a closed curve is zero.

The singularities of $1/(z^2 + 4)$ are $z = \pm 2i$, both with $|z| = 2 > 1$, i.e. outside C.

The integrand is analytic inside C, so the integral is **0**: **(D)**.

Exam tip: Always locate the poles first; only those inside the contour contribute residues.

Q8 **Answer: 0.7552 (accept 0.7550 to 0.7554)**

Concept: $x_1 = x_0 - f(x_0)/f'(x_0)$ with $f(x) = x - \cos x$, $f'(x) = 1 + \sin x$.

$f(0.5) = 0.5 - 0.87758 = -0.37758$; $f'(0.5) = 1 + 0.47943 = 1.47943$

$x_1 = 0.5 + 0.37758/1.47943 =$ **0.7552** (the root is 0.7391)

Exam tip: Keep the calculator in radian mode for trigonometric equations.

Q9 **Answer: 49.5 (accept 49.4 to 49.6)**

Concept: Simpson's 3/8 rule: $I = (3h/8)[f_0 + 3f_1 + 3f_2 + f_3]$; it is exact for cubics only.

$f = 0, 1, 16, 81$ at $x = 0, 1, 2, 3$

$I = (3/8)(0 + 3 + 48 + 81) = (3/8)(132) =$ **49.5** (exact value $3^5/5 = 48.6$)

Exam tip: Simpson's 3/8 rule needs the number of strips to be a multiple of 3; the 1/3 rule needs an even number.

Q10 **Answer: (C) Kaplan turbine**

Concept: Part-load efficiency depends on how well the blade angles match the changed velocity triangles.

In a Kaplan turbine both the guide vanes and the runner blades are adjustable (double regulation), so the efficiency curve is flat: **(C)**.

A fixed-blade propeller turbine has a sharply peaked efficiency curve; a Francis turbine adjusts only its guide (wicket) gates.

Exam tip: Kaplan turbines suit low heads (about 2–70 m) with large, variable flows, e.g. run-of-river plants.

Q11 **Answer: 575.5 (accept 572.6 to 578.4)**

Concept: Steady-flow energy equation for an adiabatic nozzle: $V_2 = \sqrt{(2\,\eta_n\,\Delta h_s)}$, with Δh in J/kg.

Actual drop $= 0.92 \times 180 = 165.6$ kJ/kg

$V_2 = \sqrt{(2 \times 165\,600)} =$ **575.5 m/s**

Exam tip: The shortcut $V = 44.72\sqrt{\Delta h}$ (Δh in kJ/kg) gives the same result.

Q12 **Answer: (A) the turbine runs faster, so the turbine and generator are smaller and cheaper**

Concept: $N_s = N\sqrt{P}/H^{5/4}$; for fixed P and H, N is proportional to N_s.

A higher speed means a smaller runner and a smaller, cheaper generator (fewer poles): **(A)**.

But high-specific-speed (Kaplan) runners have high relative velocities and low pressures at exit, so they are more prone to cavitation and need a lower setting.

Exam tip: Thoma's critical cavitation factor rises steeply with specific speed.

Q13 **Answer: (D) C**

Concept: Mild steel shows a sharp yield point: the stress drops from the upper yield point and then fluctuates about a lower value while Lüders bands spread.

A = proportional limit, B = upper yield point, C = lower yield point, D = ultimate tensile stress, E = fracture (engineering stress falls after necking).

The lower yield point C is taken as the yield strength in design: **(D)**.

Exam tip: The yield-point phenomenon is caused by carbon and nitrogen atoms pinning dislocations (Cottrell atmospheres).

Q14 **Answer: 95.94 (accept 95.45 to 96.43)**

Concept: A hypereutectoid steel consists of pearlite plus proeutectoid cementite (at the grain boundaries); use the lever rule just below 727 °C.

Pearlite = (6.67 − 1.0)/(6.67 − 0.76) = 5.67/5.91

Pearlite = **95.94 %**, proeutectoid cementite = 4.06 %

Exam tip: The grain-boundary cementite network makes slowly cooled hypereutectoid steels brittle; they are spheroidised before use.

Q15 **Answer: (C) sub-zero (cryogenic) treatment immediately after quenching**

Concept: As carbon and alloy content rise, M_f falls below room temperature, so some austenite remains untransformed after quenching.

Cooling the steel to −80 °C or below (sub-zero treatment) takes it past M_f and converts most of the remaining austenite into martensite: **(C)**.

A higher austenitising temperature dissolves more carbon and Ni/Mn lower M_s, both *increasing* retained austenite.

Exam tip: Gauges and bearings get sub-zero treatment for dimensional stability.

What next?

Day 54 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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